

What Does AQI 180 Feel Like? Co-Designing Personal Air Filters for Environmental Sensemaking

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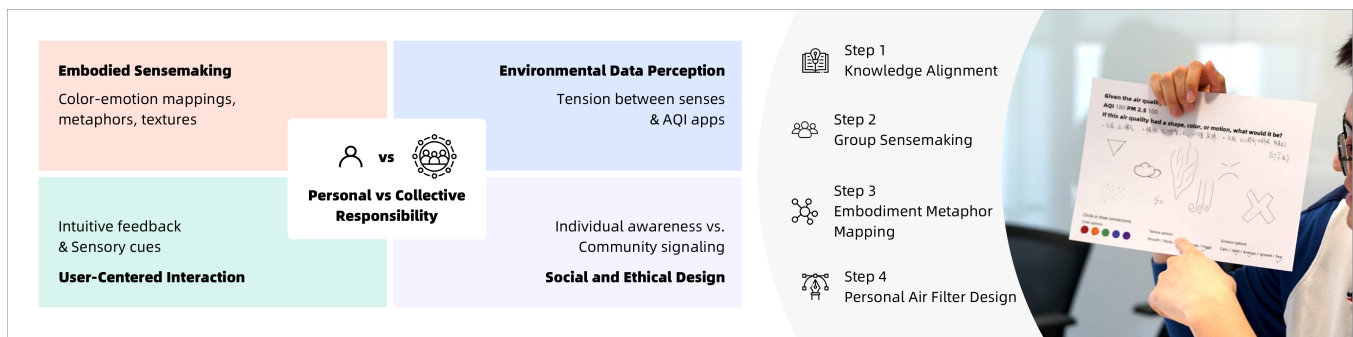


Figure 1: Conceptual framing of embodied environmental sensemaking and co-design workflow.

Abstract

Air quality is often conveyed through abstract metrics like AQI and $PM_{2.5}$, but these numbers don't capture the everyday experience of polluted air. In a co-design workshop with 15 participants, we explored how people interpret air quality using embodied metaphors and how this informs personal air filter design. Participants associated clean air with smooth, cool, calming sensations, while polluted air was described as sharp, heavy, and alarming. Using these mappings, they created air filter devices that translated air quality data into intuitive sensory feedback, such as color, vibration, and ambient cues. Our findings show embodied metaphors can improve environmental sensemaking beyond numbers and highlight tensions between personal protection and collective responsibility. We discuss implications for environmental HCI and participatory approaches linking personal experience to broader social and environmental goals.

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CCS Concepts

• Human-centered computing → Empirical studies in HCI; Participatory design.

Keywords

Environmental HCI; Air Quality Index (AQI); Environmental sensemaking; Co-design; Embodied metaphors; Data physicalization; Personal air filters; Sustainability

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1 Introduction

Air pollution is predominantly communicated through abstract metrics, such as the Air Quality Index (AQI) and particulate matter concentrations (e.g., $PM_{2.5}$). Yet for people living under smoggy skies, these numbers rarely translate into an intuitive understanding of how polluted air is experienced in everyday life. An AQI value like 180 may indicate “unhealthy” conditions, but what does it actually feel like to breathe such air? Without a clear experiential anchor,



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numerical air quality information can remain distant, difficult to interpret, and easy to ignore in daily decision-making.

In this work, we explore how individuals make sense of air quality data through embodied metaphors and how these interpretations inform the design of personal air filter devices. We report on a co-design workshop in which participants were invited to translate air quality conditions into sensory qualities—such as colors, textures, and emotions—and to prototype personal air filters that communicate pollution levels through intuitive, bodily perceivable feedback. By focusing on participants’ lived experiences and self-generated metaphors, we examine how abstract environmental data can be grounded in perception and action rather than treated as purely informational signals. Prior work shows that coupling environmental data with lived experience can enhance sensemaking, motivating participatory and embodied design approaches.

This paper presents a concise case study of co-designing for environmental sensemaking. Our contributions are threefold: (1) empirical insights into how people experientially interpret AQI and $PM_{2.5}$ levels through embodied metaphors (e.g., smooth versus sharp, calming versus alarming), (2) a set of design dimensions for personal air filter devices that translate air quality data into sensory feedback, and (3) reflections on the tension between personal protection and collective responsibility in sustainability-oriented design. Together, these contributions highlight opportunities for environmental HCI to move beyond numeric displays toward experiential, human-centered representations of air quality that support both individual awareness and broader environmental engagement. Figure 1 illustrates our conceptual framing of embodied environmental sensemaking and the co-design workflow, highlighting how abstract air quality data is translated into experiential metaphors and personal device concepts.

2 Related Work

2.1 Environmental Data Sensemaking in HCI

HCI researchers have long sought to make environmental data, such as air quality measurements, more interpretable and actionable for everyday people. Prior work shows that simply providing numeric data often fails to engage users; instead, integrating data with narratives or familiar experiences can enhance understanding[10, 11]. The study "Making Air Quality Data Meaningful" exemplifies this trend by integrating objective sensor readings with personal narratives, effectively bridging the gap between environmental data and lived experience[11]. This approach aligns with the concept of environmental sensemaking, in which data is contextualized within human experiences to inform decision-making[4, 7, 12]. Further, narrative visualization and participatory storytelling in HCI research—such as participatory data narration—highlight how personal and communal experiences can enhance the interpretability of complex environmental data[9].

2.2 Co-Design and Embodied Metaphors

Co-design has proven valuable for generating metaphors and design ideas that resonate with target users’ experiences. Ghorayeb et al. found that involving older adults as co-designers yielded visual metaphors grounded in their own life experiences, leading to more accessible and meaningful interfaces[8]. Similarly, Cajamarca

et al. demonstrated that participatory design with older adults in Chile and Ecuador led to co-created visual representations of health data that reflected users’ lived experiences rather than abstract metrics[2]. In general, end users are often ideal co-designers for creating representations of data that avoid technical jargon and instead connect to everyday sensations and values[1, 3]. This insight motivates our approach: rather than imposing designer-chosen visuals for air quality, we invited participants to co-create mappings (e.g., what color or feeling corresponds to “healthy air” vs. “hazardous air”), building on participatory design principles shown to empower users and ground visual metaphors in their sensory and cultural understandings[13, 14].

2.3 Personal vs. Collective Responsibility in Sustainable Design

Designing for sustainability often raises questions about the locus of responsibility: Should solutions target individual behavior change, or systemic, collective action? This debate is well-established in HCI’s sustainability discourse. For example, Dourish critically examined how sustainability design often focuses too narrowly on individual responsibility, arguing instead for attention to collective and political dimensions[5]. Similarly, Dove et al. noted that while personal sensing technologies (e.g., monitoring one’s air quality exposure) may empower users, they risk obscuring the systemic and infrastructural roots of environmental problems[6].

Our study addresses this tension by examining how participants conceptualize their personal air filter devices within the broader context. Some envisioned devices that simply help themselves (personal responsibility), while others imagined signaling devices or data-sharing features that contribute to collective awareness. This duality resonates with ESG principles: environmental health is a shared burden, and technological designs can either silo the problem to individuals or help galvanize community and organizational responses. In the Discussion, we reflect on how co-designed solutions can balance personal and collective responsibility for environmental well-being.

3 Methodology

3.1 Workshop Design and Participants

We conducted a co-design workshop with 15 participants (8 female, 7 male; ages 22-36, diverse backgrounds) in January 2026. Participants were recruited through local community boards and an environmental advocacy group in our city. Many had personal experiences with poor air quality—such as asthma, commuting in traffic, or living through seasonal wildfires—ensuring they could draw from lived experience when discussing AQI and pollution. The workshop was a 2-hour in-person session at a community center, facilitated by the authors with ethics approval from our institution.

At the start, we introduced basic facts about air quality (for instance, explaining that AQI 180 falls in the “Unhealthy” category, while AQI 40 is considered “Good”). Rather than focusing on technical definitions, we framed the workshop with the guiding question: “What does AQI 180 feel like, and how might we design a personal device to help you respond to it?” This question set the tone for imaginative, experience-based design. The structure, sequence, and

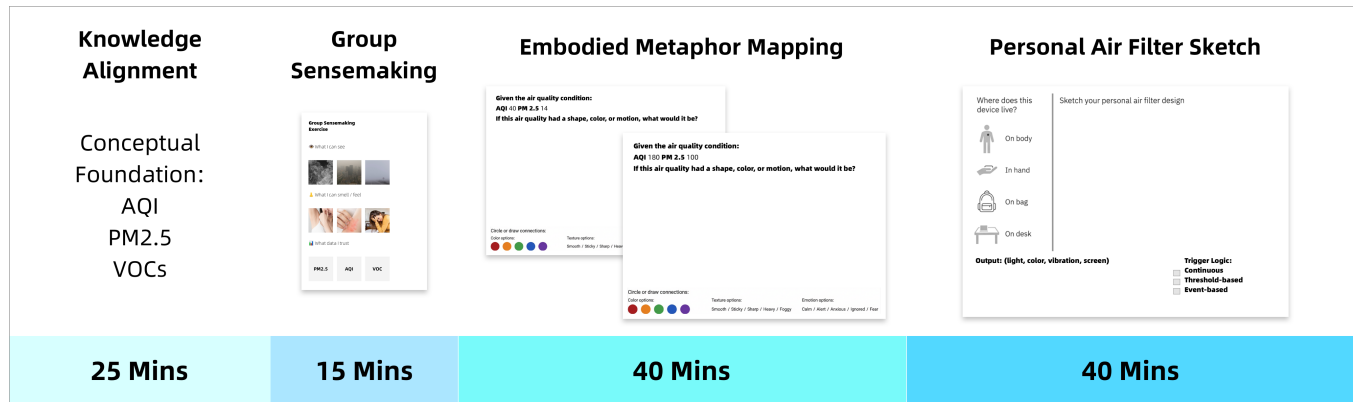


Figure 2: Structure and timing of the co-design workshop activities.

timing of the co-design workshop activities are summarized in Fig. 2.

3.2 Activities and Data Collection

The workshop began with a group sensemaking exercise, where participants listed and discussed cues they use to judge air quality—what they see, smell, feel, and the data sources they trust. This revealed that some rely on personal senses (“if I smell smoke, I don’t need a number”), while others trust quantitative data (“I check AQI on my phone every morning”). The group also explored how their device ideas might combine these cues, such as integrating odor detection and numeric displays. Next, participants used worksheets to map air quality metaphors, describing good and bad air days with colors, textures, and emotions. Finally, each person sketched a concept for a personal air filter, considering where it would be worn, how it would provide feedback (color, vibration, sound), and when it would alert (continuous, threshold, or event-based). Designs and notes were collected for analysis.

Throughout the workshop, researchers circulated to observe and ask probing questions. We collected the completed worksheets, sketches, and sticky notes from the exercises. The session concluded with each participant briefly sharing their device concept with the group, which allowed additional insights and commentary to emerge in a plenary discussion.

4 Analysis

We conducted a qualitative analysis of workshop artifacts and transcripts. Two researchers independently coded worksheets, sketches, and group discussions, identifying themes and unique ideas. We tallied descriptors for air quality metaphors, categorized device types and outputs, and captured key sensemaking and responsibility themes. Findings were synthesized into concise results, illustrated with edited participant quotes and design examples. Despite a small dataset, the multi-method approach offered rich qualitative insight, allowing us to highlight both common patterns and diverse responses. Rather than aiming for generalizable claims, our analysis focuses on identifying recurring patterns and design-relevant tensions that can inform future environmental HCI research.

5 Findings

5.1 Embodied Interpretations of Air Quality (AQI)

Participants consistently interpreted air quality through embodied metaphors rather than numeric reasoning. Clean air (AQI 40) was associated with smooth, cool, and calming sensations, often described using light or natural colors such as blue and green. These conditions were perceived as unobtrusive and unremarkable, blending into the background of everyday experience. Figure 3 shows representative embodied metaphor mappings and personal air filter sketches produced during the workshop, illustrating how participants translated AQI levels into sensory and design representations.

In contrast, polluted air (AQI 180) was described using sharp, heavy, and oppressive sensations, frequently mapped to red or dark hues signaling danger or alarm. Participants associated poor air quality with physical discomfort and emotional unease, framing pollution as something felt directly in the body rather than understood abstractly. Across participants, numeric AQI values were intuitively translated into sensory and affective terms, suggesting that people naturally rely on embodied cues to make sense of invisible environmental conditions. Selected artifacts from this analysis are presented in Fig. 3 to ground the reported themes in participants’ design work.

5.2 Personal Air Filter Device Concepts

Participants’ air filter designs reflected how embodied interpretations of air quality informed concrete interaction strategies. Rather than converging on a single solution, the concepts clustered along three design dimensions: form factor, output modality, and trigger logic.

- **Form Factors:** Designs spanned wearable, portable, and ambient forms. Wearable devices emphasized continuous bodily awareness and discretion, portable devices supported situational checking and flexibility, and ambient devices framed air quality as a shared property of spaces. These variations reflect different assumptions about where environmental awareness should reside—on the body, with the user, or in the surrounding environment.

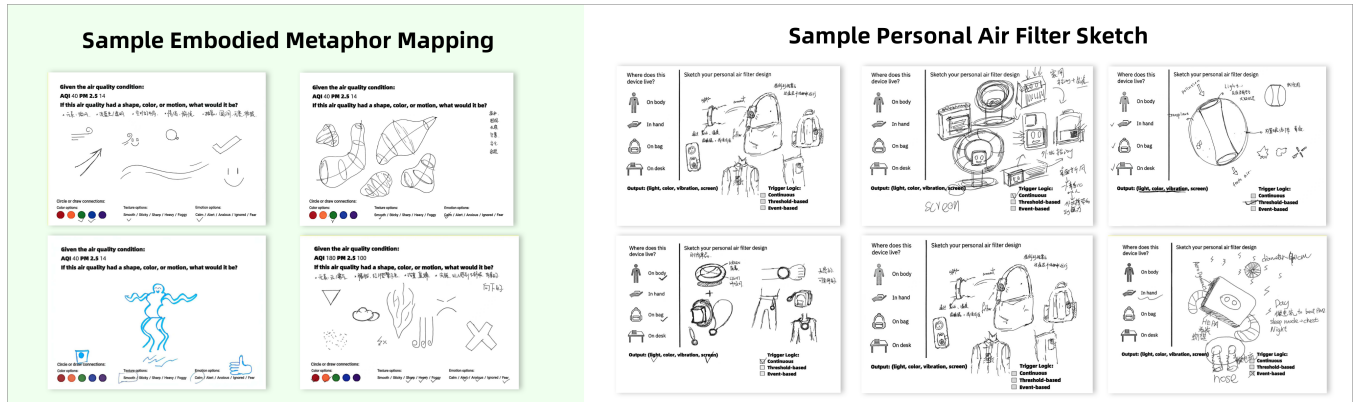


Figure 3: Representative embodied metaphor mapping and personal air filter design sketches produced during the co-design workshop

- **Output Modalities:** Visual feedback was the dominant modality, typically using color to signal air quality states. Haptic cues such as vibration were commonly used for subtle alerts, while audio feedback appeared sparingly and was reserved for urgent conditions. Overall, participants favored glanceable, ambient cues over detailed numeric displays, prioritizing intuitive awareness rather than precise monitoring.
- **Trigger Logic:** Devices varied in how feedback was activated. Some designs emphasized continuous feedback, others relied on threshold-based alerts, and a smaller subset proposed context- or event-based activation. These strategies reflect trade-offs between constant awareness and selective notification, highlighting the importance of adaptable systems that accommodate different sensitivities and lifestyles.

6 Discussion

Our findings illustrate how co-design can uncover rich, embodied understandings of environmental data and translate them into design solutions. Here we reflect on the implications of this work for HCI and design research:

Embodied Metaphors as Design Material: Participants didn't view AQI 180 as just a number—they experienced it as color, texture, and emotion. This highlights the value of using sensory metaphors in environmental informatics design. Rather than relying solely on technical gauges, designers can incorporate metaphoric representations, like a device that glows red or feels sharp when pollution is high, matching users' intuitive language. This multisensory, affective approach makes data more meaningful and connects to data physicalization: translating information into experiences people can feel or see. Our study, though small, revealed common metaphors (smooth vs sharp, green vs red, calm vs fear) across participants. Future designs could allow users to personalize feedback to their unique associations—such as choosing between a flashing light or a vibration for alerts. Co-design is well-suited for uncovering these personal mappings, supporting more resonant, user-centered solutions.

Design for Reflection, not just Reaction: Rather than directly prescribing actions, participants' designs emphasized ongoing awareness that supports reflection and situational decision-making. By translating air quality into perceptible cues, these systems can help users notice temporal and contextual patterns (e.g., recurring pollution at specific times or locations), which in turn may inform everyday choices such as route selection, outdoor activity timing, or protective behaviors. This reflects the "slow technology" concept in HCI: supporting reflection rather than constant reaction. Participants preferred gentle, informative systems over alarm overload, desiring tools that fit seamlessly into daily life. Designers should balance urgency with calm awareness, using culturally relevant cues to nudge without causing anxiety or fatigue. In environmental HCI, empowering people means fostering awareness and agency without overwhelming them in the face of global challenges.

Personal Devices Within a Broader Sustainability Context: While personal air quality devices can empower individuals to protect themselves and stay informed, they also raise concerns about shifting responsibility away from collective and systemic solutions. The findings highlight the importance of designing personal technologies that acknowledge this tension, creating opportunities to connect individual experience with shared awareness, community engagement, or broader sustainability efforts rather than treating pollution as solely a personal problem.

7 Limitations

As an exploratory co-design study with 15 participants, this work does not aim for statistical generalization, but instead surfaces generative patterns and design considerations grounded in lived experience. We also focused on AQI and $PM_{2.5}$, but other pollutants or contexts—like ozone, VOCs, indoor air, or wildfire events—could inspire different metaphors or design needs. Despite these limitations, engaging people in co-design for environmental sensemaking delivers creative solutions and critical reflections that enrich HCI's approach to sustainability.

8 Conclusion

We presented “What Does AQI 180 Feel Like?”, in which 15 participants reimagined air quality data using color, texture, and emotion, then designed personal air filter devices for environmental sensemaking. Participants mapped good air to green, smooth, calm and bad air to red, sharp, fear, revealing an intuitive human lexicon for pollution. Using these metaphors, they proposed devices that make pollution palpable—such as glowing filters or vibrating warnings—highlighting opportunities for HCI to create more embodied, intuitive interfaces. Our findings emphasize that while personal devices empower users and raise awareness, they must be integrated into broader, collective systems for real impact.

We hope this work inspires further research connecting co-design, data embodiment, and sustainability. Involving people in answering “what does it feel like?” ensures more resonant technologies and fosters sensemaking and agency. As a poster contribution, this work is intended to provoke discussion and inspire further investigation rather than to provide definitive evaluations. Personal insights can spark broader change, and human-centered, metaphor-rich approaches can advance sustainability and collective well-being.

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